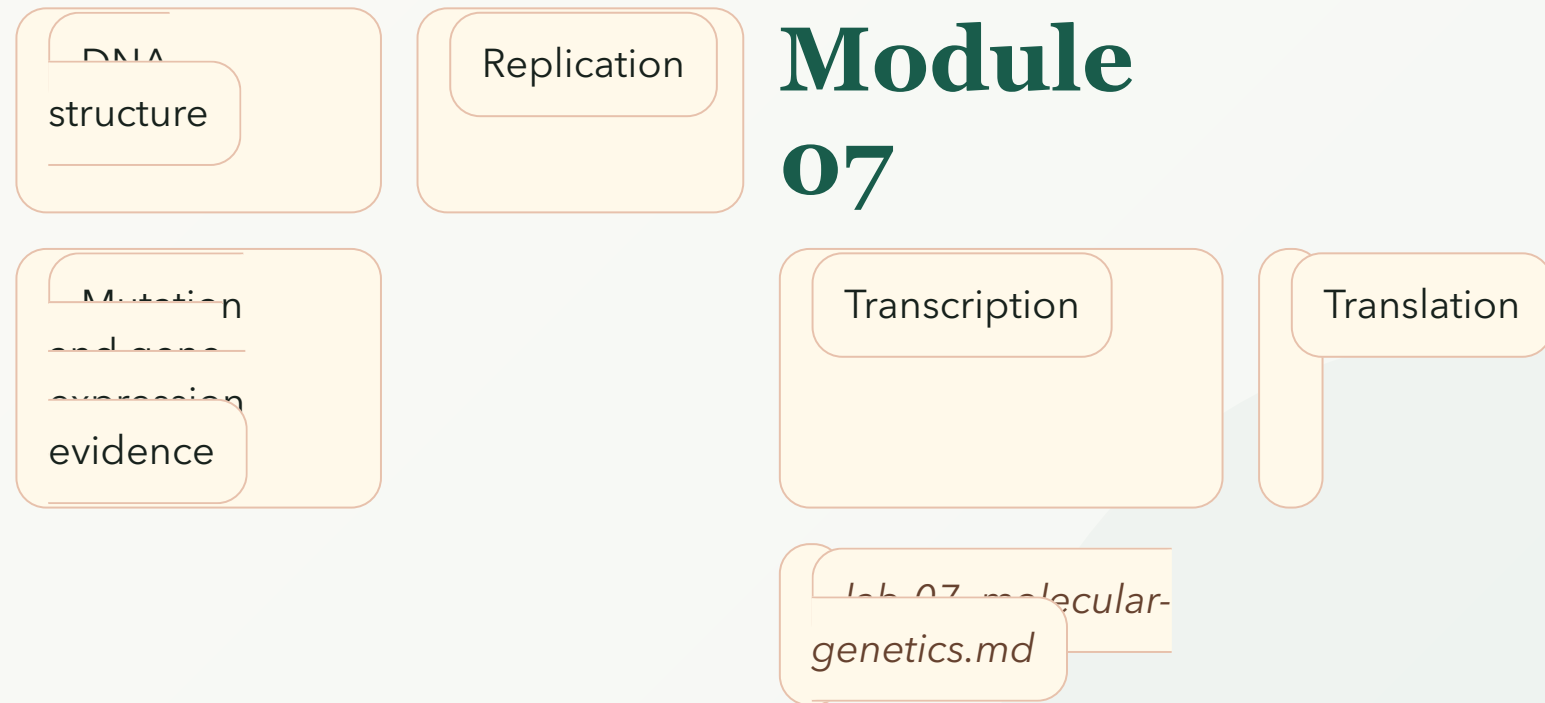


MOLECULAR GENETICS

Module map

- Module 07: Molecular Genetics
- Connected lab: lab-07_molecular-genetics.md
- Use this deck as the visual path through the module's evidence and retrieval work.



MOLECULAR GENETICS

Learning objectives

- Describe DNA structure and base-pairing rules.
- Explain why DNA replication is semiconservative.
- Compare transcription and translation.
- Use codons to connect mRNA sequence to amino acid sequence.
- Explain how mutation can alter protein and trait outcomes.

OBJECTIVE 1

Describe DNA structure and base-pairing rules.

OBJECTIVE 2

Explain why DNA replication is semiconservative.

OBJECTIVE 3

Compare transcription and translation.

OBJECTIVE 4

Use codons to connect mRNA sequence to amino acid sequence.

OBJECTIVE 5

Explain how mutation can alter protein and trait outcomes.

MOLECULAR GENETICS

Topic sequence

- DNA structure: DNA stores information in nucleotide sequences and complementary base pairing.
- Replication: Replication copies DNA using each strand as a template.
- Transcription: Transcription produces RNA from a DNA gene.
- Translation: Translation uses mRNA codons and ribosomes to build proteins.
- Mutation and gene expression evidence: Mutations and expression patterns connect molecular changes to traits.

TOPIC 1
DNA structure

TOPIC 2
Replication

TOPIC 3
Transcription

TOPIC 4
Translation

TOPIC 5
Mutation and gene expression evidence

MOLECULAR GENETICS

Module 07: Molecular Genetics Evidence Map

- Module focus: Molecular Genetics
- Central claim: Molecular Genetics explains DNA information flow from sequence to RNA, protein, and trait evidence.
- Key nodes to connect: DNA structure, Replication, Lab evidence
- Reasoning move: use 'requires' to explain relationships, not memorize terms.
- Lab connection: lab-07_molecular-genetics.md supplies an evidence surface for the map.

Teaching note: Ask students to explain one edge in their own words before moving on.

MOLECULAR GENETICS

Module 07: Molecular Genetics Reasoning Sequence

- Module focus: Molecular Genetics
- Trace the model: DNA structure -> Replication -> Transcription -> Translation
- Inputs become outputs: DNA structure, Replication -> Describe DNA structure and base-pairing rules., Explain why DNA replication is semiconservative.
- Feedback check: lab-07_molecular-genetics.md evidence checks whether students can explain transcription.
- Lab connection: lab-07_molecular-genetics.md gives students a process to observe or test.

Teaching note: Emphasize where the process can fail, slow down, or be revised by evidence.

MOLECULAR GENETICS

Terms as evidence handles

- DNA: Nucleic acid that stores hereditary information.
- Nucleotide: DNA or RNA subunit with sugar, phosphate, and base.
- Replication: Process that copies DNA.
- Transcription: Synthesis of RNA from a DNA template.
- Translation: Synthesis of protein from mRNA codons.
- Codon: Three-base mRNA sequence read during translation.

DNA

Nucleic acid that stores hereditary information.

NUCLEOTIDE

DNA or RNA subunit with sugar, phosphate, and base.

REPLICATION

Process that copies DNA.

TRANSCRIPTION

Synthesis of RNA from a DNA template.

TRANSLATION

Synthesis of protein from mRNA codons.

CODON

Three-base mRNA sequence read during translation.

MOLECULAR GENETICS

Lab connection

- Lab file: lab-07_molecular-genetics.md
- Lab evidence should test or illustrate: Transcription
- Students should leave with a concrete observation, comparison, or model tied to the module claim.

QUESTION

DNA structure

EVIDENCE

lab-07_molecular-genetics.md

CLAIM

Describe DNA structure and base-pairing rules.

MOLECULAR GENETICS

Contrast check

- Do not stop at: DNA stores information in nucleotide sequences and complementary base pairing.
- Stronger explanation: Mutations and expression patterns connect molecular changes to traits.
- The goal is to move from a named idea to a testable biological explanation.

SURFACE

DNA stores information in nucleotide sequences and complementary base pairing.

DEEPER

Mutations and expression patterns connect molecular changes to traits.

MOLECULAR GENETICS

Module 07: Molecular Genetics Retrieval and Lab Check

- Module focus: Molecular Genetics
- Retrieval prompt: How does DNA structure store information?
- Answer check: Describe DNA structure and base-pairing rules.
- Required terms: DNA, Nucleotide, Replication
- Lab connection: lab-07_molecular-genetics.md; lab-07_molecular-genetics.md supplies evidence for molecular genetics evidence from sequence and expression reasoning.

Teaching note: Pause here. Students answer first without notes, then compare to the checks.

MOLECULAR GENETICS

Practice quiz bridge

- 1. Which statement best defines DNA in Module 07?
- 2. Which learning goal best supports the topic "Replication"?
- 3. How should lab-07_molecular-genetics.md support Module 07?
- 4. Which retrieval move best prepares a student for Molecular Genetics?

QUIZ 1

A

QUIZ 2

B

QUIZ 3

C

QUIZ 4

D

MOLECULAR GENETICS

Synthesis and exit ticket

- Exit claim: explain dna structure using evidence from lab-07_molecular-genetics.md.
- Must include: DNA, Nucleotide, Replication.
- Revision prompt: what evidence would change your explanation?

CLAIM
DNA structure

EVIDENCE
lab-07_molecular-genetics.md

REVISION
What would change your mind?